KOREA UNIVERSITY
Business School

Seminar 2 — Principles in Data Science

BUSS256: Introduction to Big Data Analytics and Interpretation

Korea University Business School
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Data literacy: can I trust this data for this question?

Data literacy = judging whether data is useful for a problem and arguing with evidence from data; it means reading and understanding data.

1

Where did it come from?

2

What does it measure?

3

Is it good enough for this question?

Algorithm literacy comes later, with the techniques.

Learning objectives

1. Use the core terms in plain business language

2. Name the four key elements of business analytics and how they connect

3. Explain what BA is used for and what it cannot promise

4. Practise one habit — before you trust a number or claim, ask how you would check it

Vocabulary test

How many terms do you know?

1. Data
2. Information
3. Data Science
4. Data Mining
5. Analytics (vs. Analysis)
6. Data Analytics
7. Business Analytics
8. Big Data
9. Business Problems
10. Descriptive/Predictive Analytics

Data and information

Data = raw facts (flour, eggs); **information** = data organized so it can answer a question (the cake).

ILLUSTRATIVE EXAMPLE

1 Data
raw facts

→

2 Information
data organized to answer a question

→

3 Decision
a decision to make

Raw transactions

Product A	Product B
Product B	Product A
Product A	Product A
Product A	Product B

This week:

Product A = 5 units

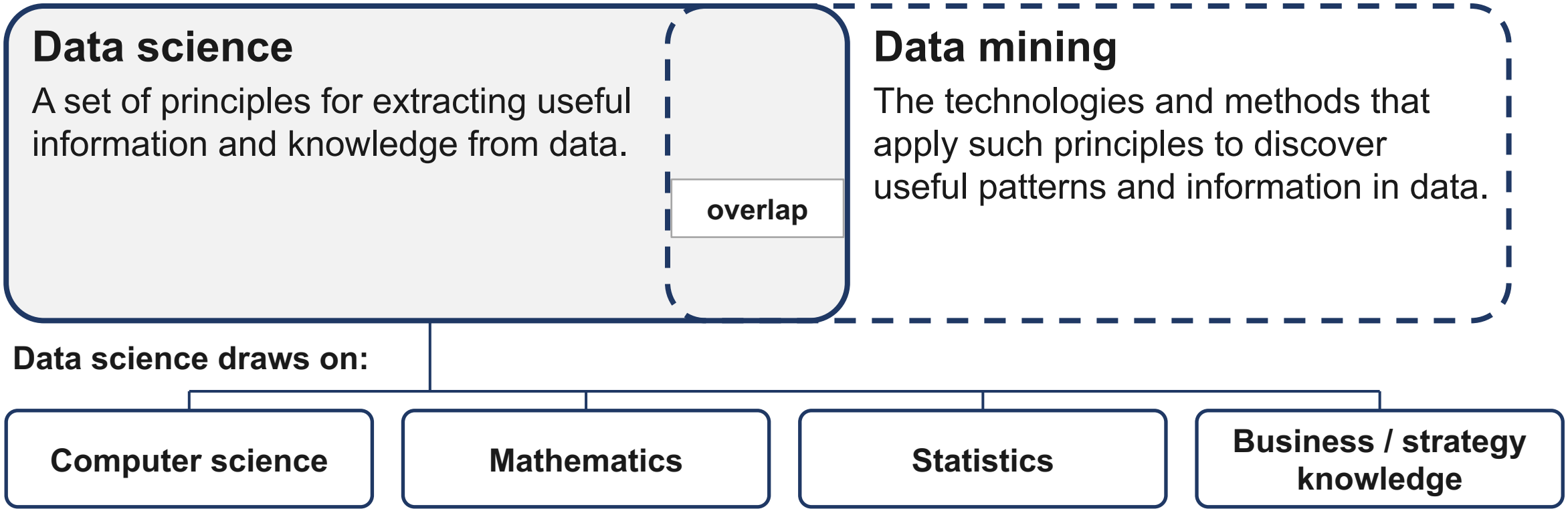
Product B = 3 units

Decision question:

“Should we stock more A next week?”

Is a spreadsheet of 10,000 transactions data or information; what turns it into information?

Data science and data mining



They overlap; one is not simply "inside" the other.

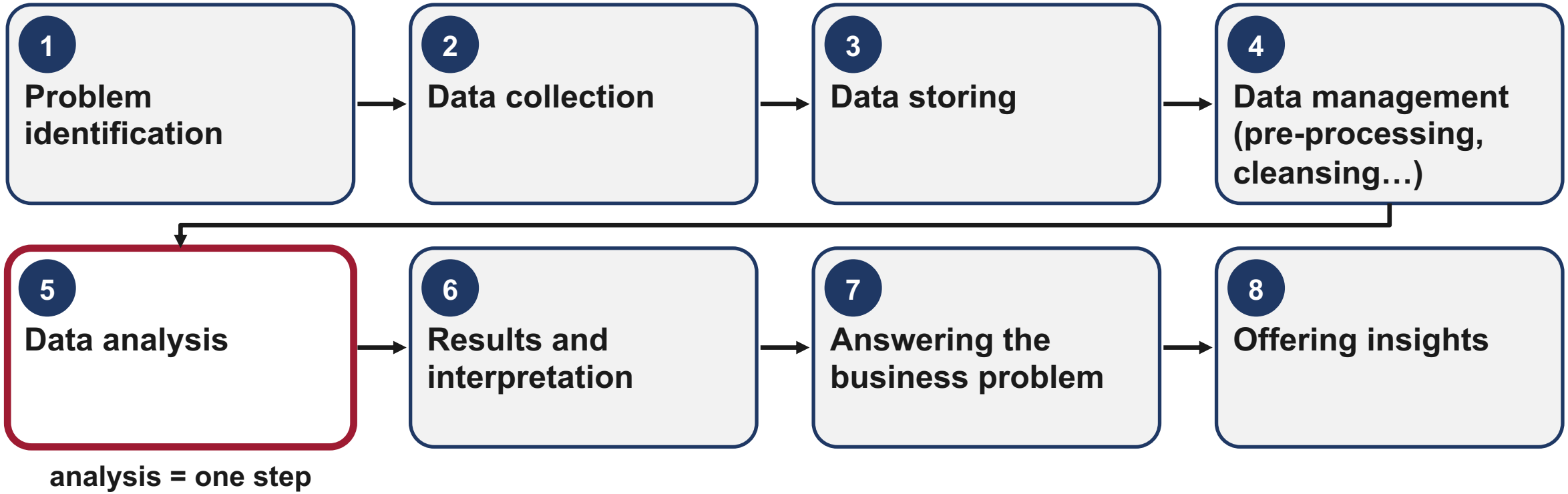
Analytics: our working definition

> Analytics is the **scientific process** of **transforming data** into **insight** for making **better decisions**. (INFORMS)

Different sources draw the boundaries differently; this is the convention used in BUSS256.

Source: INFORMS, “Operations Research & Analytics” and “FAQs About O.R. & Analytics” (informs.org).

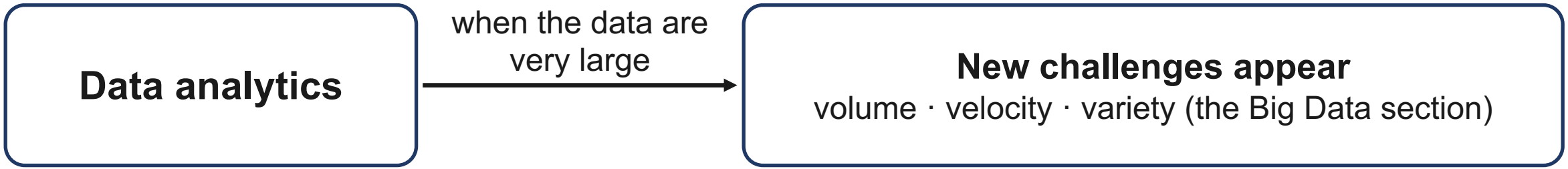
The analytics process



Analysis is one step within the broader analytics process; analytics may include descriptive, predictive and decision-oriented work.

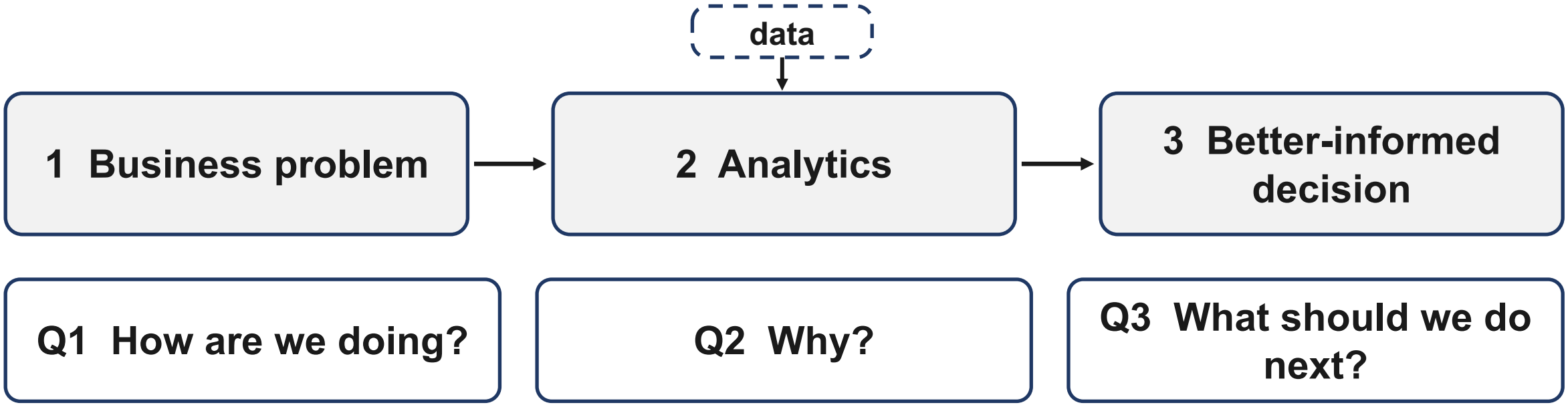
Data analytics

Data analytics is analytics focused on analyzing data — collecting, organizing and examining data to find patterns and useful information.

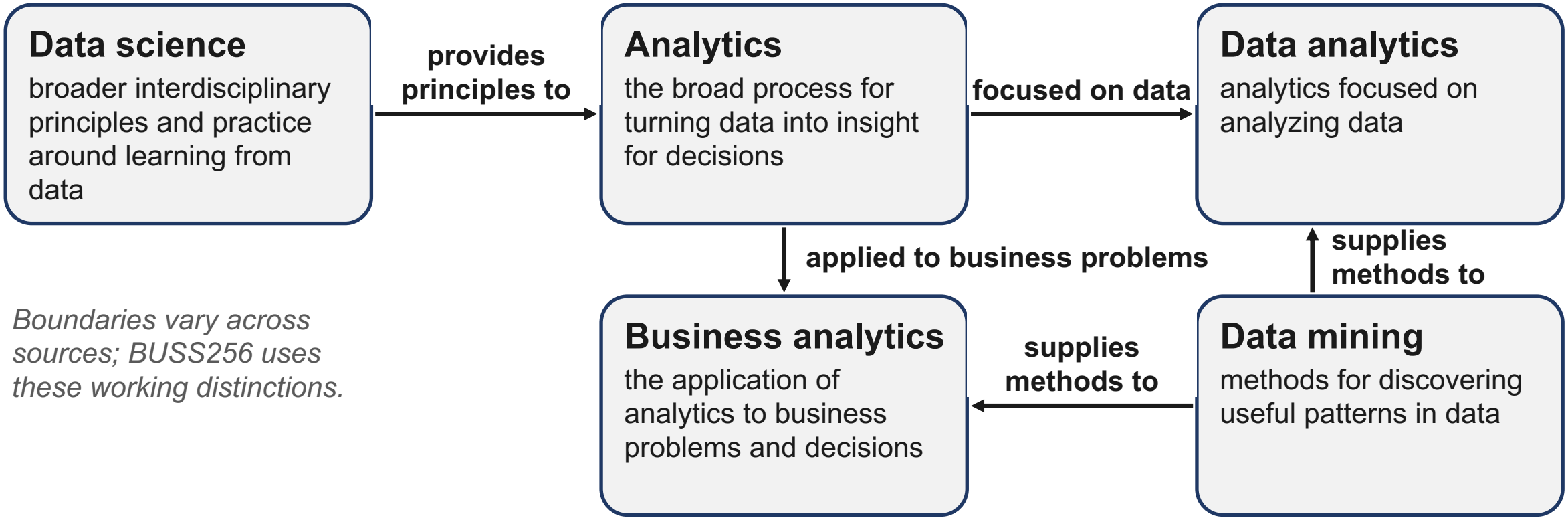


Business analytics: from a business problem to a better-informed decision

Business analytics applies data and analytics to a business problem to generate insight for decisions. It helps answer three questions:



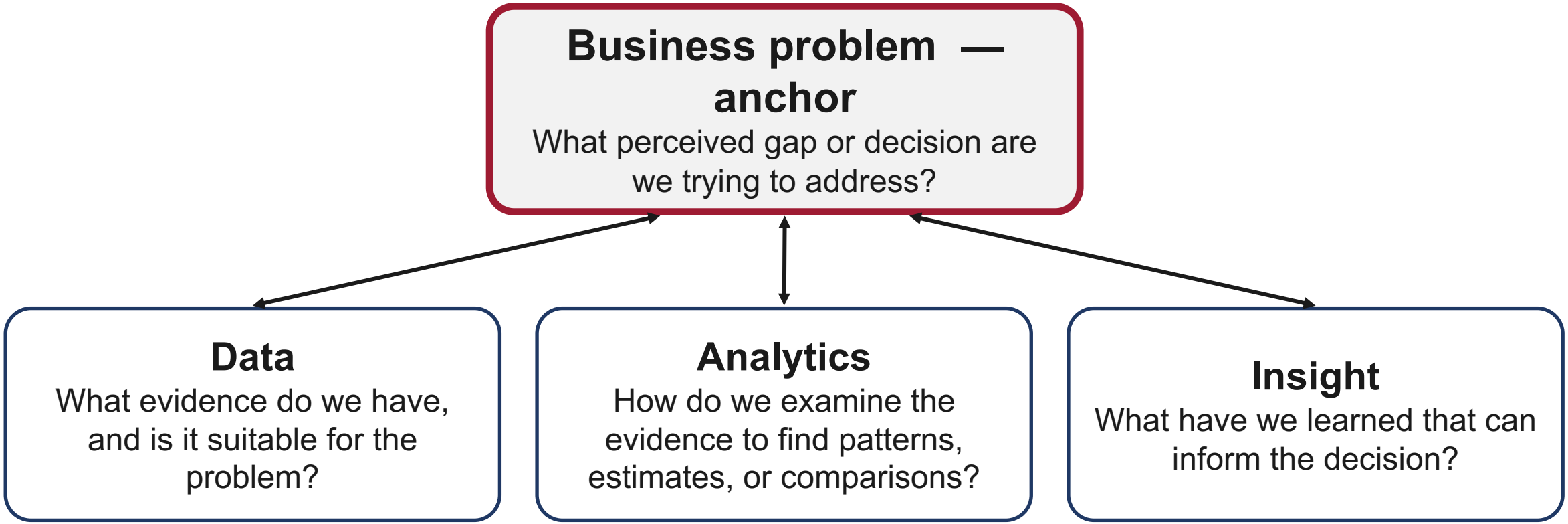
Part 1 recap: vocabulary and relationships



Boundaries vary across sources; BUSS256 uses these working distinctions.

Which terms from the opening vocabulary test can you now explain in your own words?

Four elements of business analytics



These are connected elements, not a rigid one-way sequence. The later insight-loop slide shows how they interact in a decision process.

Big data: data from everywhere

Data comes from web and e-commerce, purchases at stores, bank and card transactions, social networks.



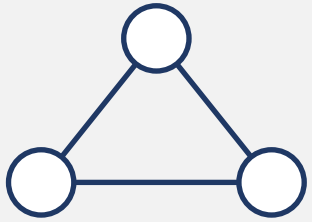
Web / e-commerce



Purchases at stores



Bank / card transactions



Social networks

Collecting more data has become easier; turning it into useful information remains hard.

More data ≠ more information

HISTORICAL EXAMPLE

What the report described

A January 2008 TechCrunch report, citing Google engineering work, described Google as processing roughly **20 PB of data per day as of September 2007** — large-scale data processing across Google's MapReduce workloads.

What it is NOT

"20 PB of user searches per day."

Historical example — not a current figure.

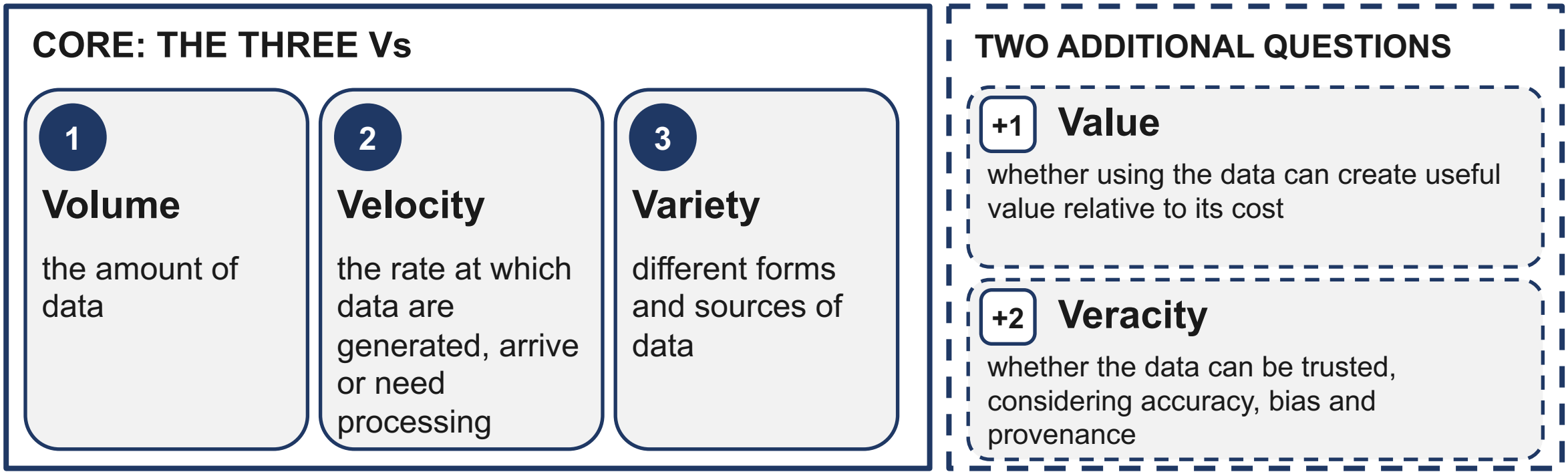
Large volume alone does not create information or insight.

Ask: **What was measured?** | **When?** | **For what purpose?**

Big data also creates special challenges in volume, velocity and variety.

Source: TechCrunch, Jan. 9, 2008 (reporting Google's processing volume as of Sept. 2007). Historical example — not a current figure.

The Vs of big data



Different frameworks use 3V, 4V or 5V labels; BUSS256 starts from the classic 3Vs and uses Value and Veracity as two additional questions.

Teaching link: check your input before you trust your output (GIGO).

Why can we collect so much data?
Two separate technology trends



Data are generated by people, homes and vehicles.

HISTORICAL EXAMPLE

COMPUTING

1965 — roughly annual doubling

1975 — revised: roughly every two years

Gordon Moore's 1965 observation projected roughly annual doubling of components (transistors) on integrated circuits; in 1975 he revised the pace to roughly every two years. An empirical observation/projection, not a natural law.

RELATED, BUT SEPARATE

STORAGE

Sept 2022 — working prototype, 2 TB microSDXC

2023 — mass production planned

Kioxia announced a working prototype of a 2 TB microSDXC card in September 2022; mass production was planned for 2023. Historical examples, not current product specifications.

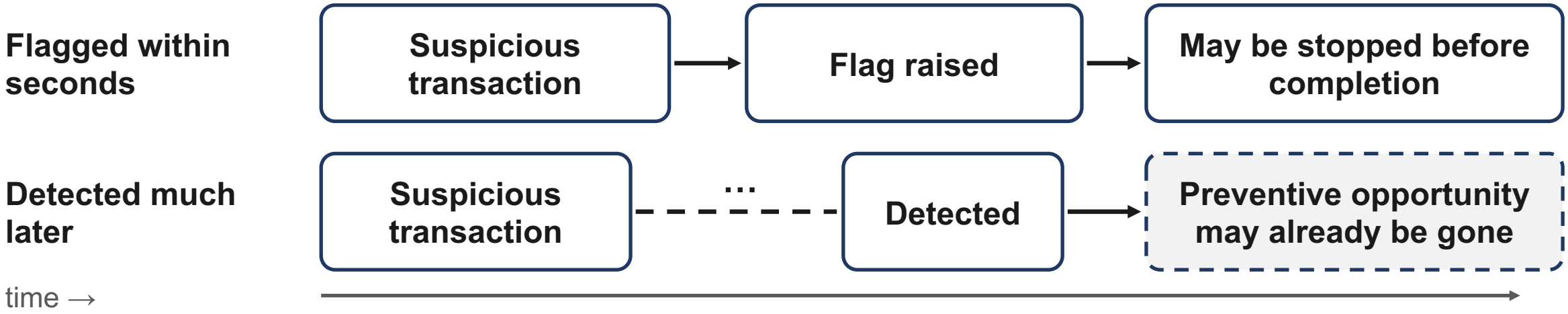
Advances in computing and storage made it increasingly feasible to collect and process much larger datasets. These are related technology trends, not evidence that one caused the other.

Sources: Intel, Moore's Law page (1965 projection; 1975 revision); Kioxia announcement, Sept. 28, 2022 (2 TB microSDXC prototype; mass production planned for 2023). Historical examples, not current product specifications.

Velocity: when timing matters

Velocity is how quickly data are generated, arrive, or need processing. Some decisions lose value when the data arrive too late.

HYPOTHETICAL EXAMPLE



Same underlying data — different decision value because of timing.

Verification prompt: “If this were a real statistic, what source and timestamp would you need before quoting it?”

Variety: structured vs. unstructured data

STRUCTURED

Organized in a predefined structure such as rows and columns (database tables, spreadsheets).

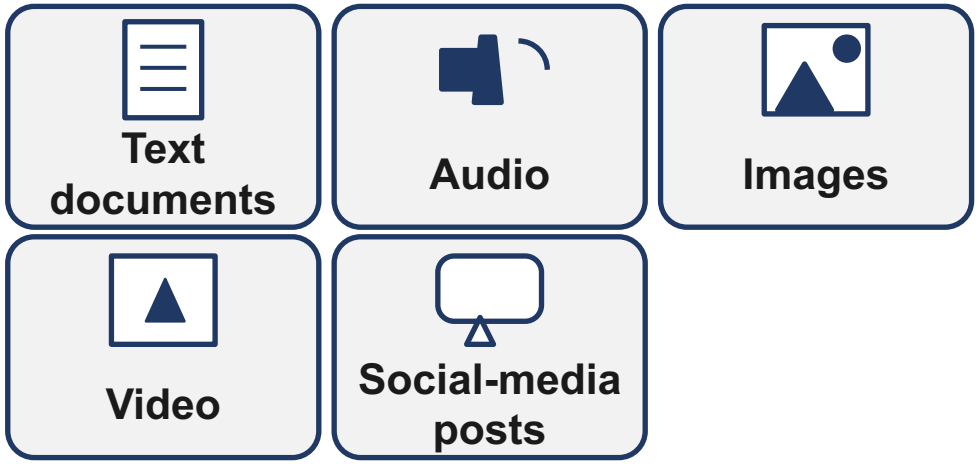
ID	Item	Amount	Date
…	…	…	…
…	…	…	…
…	…	…	…

rows and columns

vs

UNSTRUCTURED

Text documents, audio, images, video, social-media posts.



Semi-structured — JSON / logs; not developed further.

Data: why does business analytics rely on it?

1. **What kinds of data?**
2. **How do we get it (data collection)?**
3. **How do we manage it?**
4. **How do we analyze it?**

Where the course goes: data types and data management are covered later; analytical methods follow; text analysis later handles unstructured text.

Data-literacy habit: say when and how a dataset was collected (for example, the App Store dataset is a historical snapshot collected in July 2013 and is labeled that way).

Analytics: methods and types

Refers to various methods and procedures in

- Mathematics (Arithmetic: Sum, Mean, SD…)
- Statistics (Prediction, Probabilities,…)
- Computer Science (Machine Learning)
- …

Types of analytics

Descriptive Analytics:
What has happened?

Predictive Analytics:
What could happen?

Descriptive analytics

Identify, classify, and count objects or events

- Money spent or number of widgets manufactured in the past time period.
- They are critical for knowing how the organization is performing—your current situation.
- For example, they let you know if you are above or below budget, or are performing up to standards.
- This category includes reports and dashboards.

Descriptive analytics summarizes past and current performance — it tells us what happened.

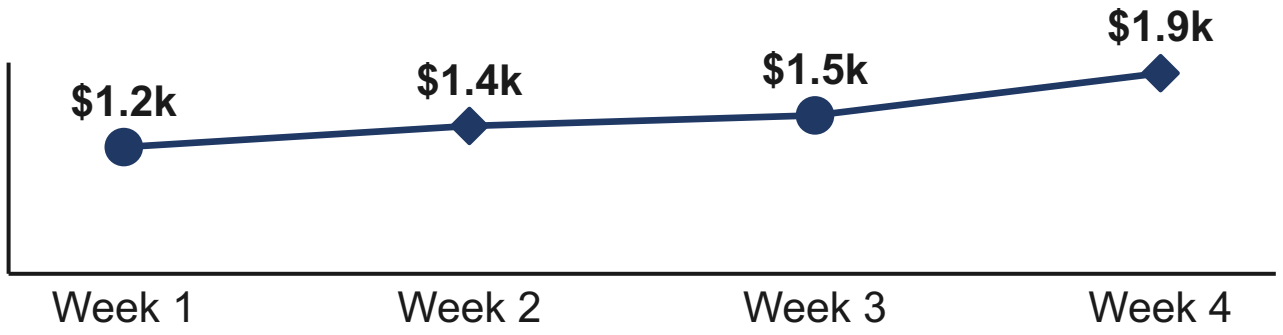
Descriptive analytics in practice: a dashboard (illustrative)

ILLUSTRATIVE EXAMPLE — Online Store

Toy values — not from a real organization or dataset

Orders	Revenue	Average order value
120	**$6,000**	**$50**

Weekly revenue



Which numbers tell us what happened?

What question would require prediction?

Predictive analytics

Look at the trend of past events to anticipate possible future outcomes.

- This can support planning by estimating possible future outcomes.
- Discover patterns and correlations in data that might be missed by the human eye.
- Build models that generate estimates under specified inputs or assumptions.



Google Flu Trends: a good model that stopped being reliable

2009 — 1

Original study (2009): GFT estimates tracked CDC-reported flu activity over the 2003–2008 study period, including a held-out validation period.

Ginsberg et al. (Nature, 2009) built GFT from search-query data.

2013 — 2

Substantial overestimation, discussed in the 2014 critique. Lazer et al. (Science, 2014): in early 2013 GFT predicted more than twice the CDC-reported proportion of influenza-like-illness doctor visits; algorithm dynamics and “big data hubris” among the problems.

2015 — 3

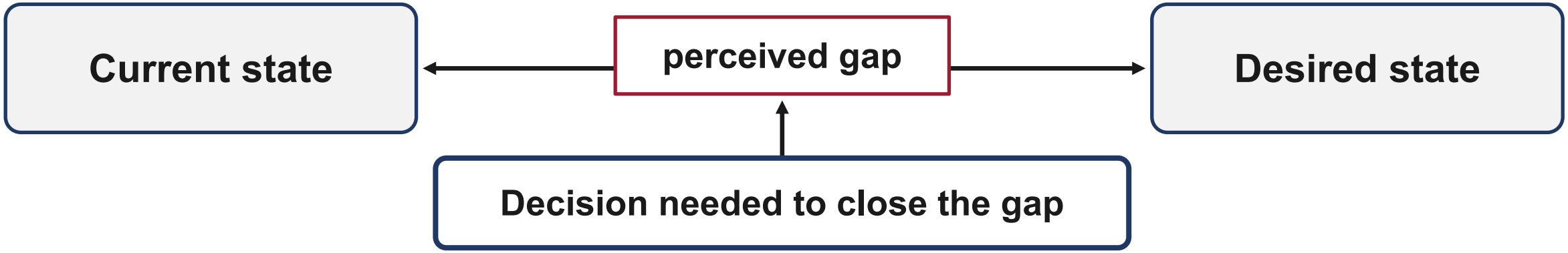
Google stopped maintaining its own Flu Trends website and provided signal data to specialist research partners (Google Research, Aug. 2015).

Lesson: A model can fit historical data and still become unreliable when the data-generating environment changes. Keep checking performance on new data.

Sources: Ginsberg et al., Nature (2009); Lazer et al., Science (2014); Google Research, “The Next Chapter for Flu Trends” (Aug. 20, 2015).

Business problems

A business problem is a perceived gap between the current state and a desired state that requires deciding how to close the gap.



Current customer churn is above the desired level	Inventory availability is below the desired service level	Sales growth is below a target

For each: what is the current state, the desired state, and the decision needed to close the gap? BA helps when the gap can be measured and the decision can be informed by data.

Common business-analytics questions — and their limits

1

How likely is client X to buy product Y?

Look at the client's purchase behaviour and patterns.

2

Which clients are at risk of going to competitors?

Look at similarities with clients who left.

3

What promotions should we offer to retain customers?

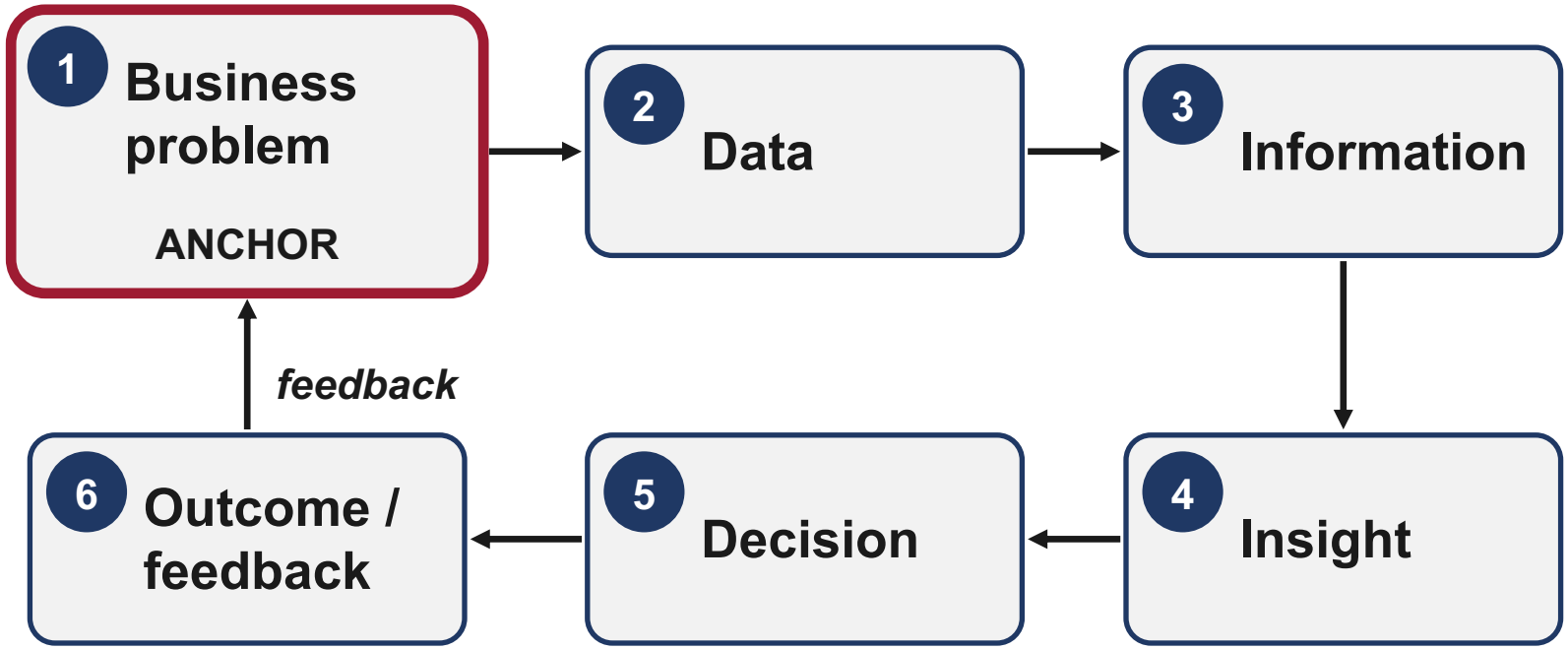
Look at relationships between offers and retention.



CAUTION — Prediction or association alone ≠ causal proof.

Prediction can identify who may be at risk; association can suggest patterns; **neither alone proves that an intervention caused an outcome.** One way to test an intervention is an A/B test.

From problem to outcome: the insight loop



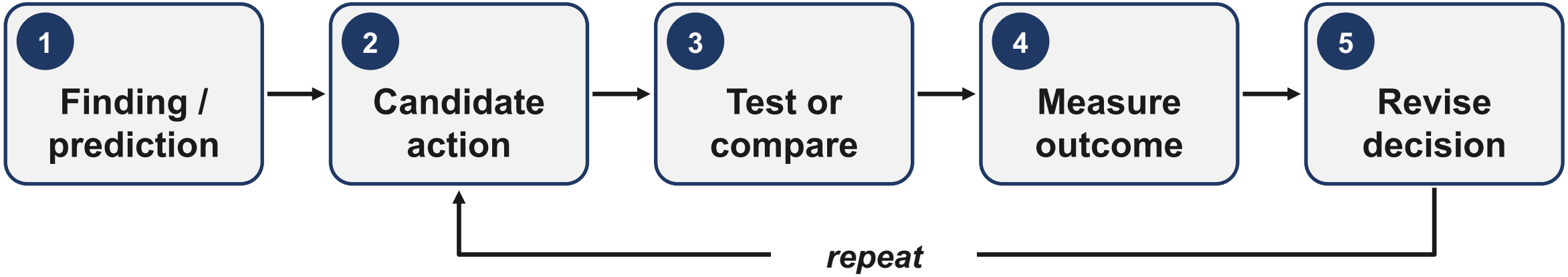
Data vs. information is on the earlier slide; this slide is about what comes after information.

Insight

An understanding of the business situation that can inform a decision. A good decision is judged by a measurable outcome consistent with the organization's goals, and it should help close the gap between the current and desired state.

From finding to action: candidate actions to test

Predictions and patterns do not automatically determine the action. Treat outputs as **candidate actions to evaluate:**



Apply this to the three questions on the earlier slide.

What business analytics can support — and what it cannot guarantee

✓ CAN SUPPORT

- Describe what happened
- Identify patterns
- Estimate likely outcomes
- Compare options
- Inform operational and strategic decisions
- Contribute to competitive advantage when used well

⚠ CANNOT GUARANTEE

- That the data are trustworthy
- That correlation is causation
- That a model that worked before will keep working
- That a recommendation will improve the outcome

The value of analytics depends on the question, the data, the method, and verification.

What's next…

Next: Data structures and basics in R

Our habit from here:

